LECTURE NOTES FOR SURV 616

Statistical Methods II

Multivariate Analysis

CLASS #2

OVERVIEW: In the last class we discussed methods of analyzing the mean from a multivariate normal distribution. In this class we examine two widely used methods of analyzing the covariance matrix of a normally distributed random vector. The first method is the method of *Principal Components* which has the two goals of data reduction, and interpretation. The second method is *Factor Analysis* which attempts to directly model the covariance matrix by assuming a data generating mechanism involving (usually) a small number of underlying *latent factors*.

Reference: Johnson and Wichern (2002). Chapters 2, 8, 9.

1. Review of Linear Algebra

We need to review a few basic results from Linear Algebra. The first two definitions give the definition of eigenvalues and eigenvectors. (Eigen is a German word for "characteristic" or "peculiar to".)

Definition 1. Let **A** be a $p \times p$ matrix and **I** be the $p \times p$ identity matrix. Then the scalars $\lambda_1,...,\lambda_p$ satisfying the polynomial equation $|\mathrm{A}\text{-}\lambda \mathrm{I}| = 0$ are called the *eigenvalues* (or *characteristic roots*) of the matrix **A**. The equation $|\mathrm{A}\text{-}\lambda \mathrm{I}| = 0$, as a function of λ, is called the *characteristic equation*.

Definition 2. Let **A** be a $p \times p$ matrix and let λ be an eigenvalue of **A**. If **x** is a *nonzero* $p \times 1$ *vector* such that $\mathrm{Ax} = \lambda \mathrm{x}$ then **x** is said to be an *eigenvector* (or *characteristic vector*) of the matrix **A** associated with the eigenvalue λ.

Example 1. Take a 2×2 matrix **A** where

$$\mathbf{A} = \begin{bmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{b} & \mathbf{c} \end{bmatrix}$$

and the determinant

$$|\mathbf{A}-\lambda\mathbf{I}| = \begin{vmatrix} \mathbf{a}-\lambda & \mathbf{b} \\ \mathbf{b} & \mathbf{c}-\lambda \end{vmatrix} = (\mathbf{a}-\lambda)(\mathbf{c}-\lambda)-\mathbf{b}^2 = \lambda^2 - \lambda(\mathbf{a}+\mathbf{c}) + (\mathbf{ac}-\mathbf{b}^2)$$

The roots of this equation are the eigenvalues:

$$\lambda = \frac{1}{2}\left[(\mathbf{a}+\mathbf{c}) \pm \sqrt{(\mathbf{a}+\mathbf{c})^2 - 4(\mathbf{ac}-\mathbf{b}^2)}\right]$$

The next result is often called the *Spectral Decomposition Theorem* (or *Singular Value Decomposition*). It allows you to write out a linear representation of a matrix in terms of its eigenvalues and eigenvectors.

Result 1. Let **A** be a $p \times p$ symmetric matrix, then **A** has p pairs of eigenvalues and eigenvectors $(\lambda_1, \mathrm{e}_1), \ldots, (\lambda_p, \mathrm{e}_p)$ where, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$, and the eigenvectors satisfy $\mathbf{e_1}'\mathbf{e_1} = \mathbf{e_2}'\mathbf{e_2} = \cdots = \mathbf{e_p}'\mathbf{e_p} = \mathbf{1}$ and $\mathbf{e_i}'\mathbf{e_j} = 0$ for $i \neq j$. We can write **A** in a *spectral decomposition* as

$$\mathrm{A} = \sum_{i=1}^{p} \lambda_i \mathbf{e_i}\mathbf{e_i}'$$

The next result gives a useful alternative interpretation of the eigenvalues and eigenvectors.

Result 2. Let **A** be a $p \times p$ symmetric matrix, where **A** has p pairs of eigenvalues and eigenvectors $(\lambda_1, \mathbf{e}_1), \ldots, (\lambda_p, \mathbf{e}_p)$ where, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. Let **x** be any $p \times 1$ vector such that $\mathrm{x}'\mathrm{x}=1$, then

$$\max_{x \neq 0} \mathbf{x}'\mathbf{A}\mathbf{x} = \lambda_1 \text{ when } \mathbf{x} = \mathbf{e}_1$$

$$\min_{x \neq 0} \mathbf{x}'\mathbf{A}\mathbf{x} = \lambda_p \text{ when } \mathbf{x} = \mathbf{e}_p$$

and

$$\max_{x \perp \mathrm{e}_1, \ldots, \mathrm{e}_k} \mathbf{x}'\mathbf{A}\mathbf{x} = \lambda_{k+1} \text{ when } \mathbf{x} = \mathbf{e}_{k+1} \text{ for } k = 1, \ldots, p-1$$

Result 3. The trace of a matrix is the sum of its diagonal elements, $tr(\mathbf{A}) = \sum_{i=1}^{p} Q_{ii}$. The trace is also the sum of the eigenvalues:

$$tr(\mathbf{A}) = \sum_{i=1}^{p} \lambda_i$$

When **A** is a covariance matrix, $tr(\mathbf{A})$ is called the *generalized variance.*

The next definition gives the definition of an *orthogonal matrix*, a term which is synonymous with *rotations* in multiple dimensions.

Definition 3. The $k \times k$ symmetric matrix **Q** is an *orthogonal matrix* if $\mathbf{QQ}' = \mathbf{Q}'\mathbf{Q} = \mathbf{I}$, or $\mathbf{Q}' = \mathbf{Q}^{-1}$.

Notice that by this definition, if the eigenvectors in Result 1 are combined together into a $p \times p$ matrix then it is an orthogonal matrix.

2. Principal Components

In SURV 615 we looked at situations where we re-expressed the predictor variables in terms of linear combinations (see notes for SURV 615, Class #5). We saw that when we used a *full rank transformation* we got exactly the same residual sum of squares and thus the same explanatory power (as measured by the residual sum of squares) as we got with the untransformed predictor variables. The application in SURV 615 was to regress somatotype (SOMA) on weights at ages 2, 9, and 18 for a sample of female teenagers. In some cases we saw that all of the untransformed predictor variables were significant in the regression, but when looking at a *full rank linear reparametrization* we saw that some of the transformed variables were not significant in the regression. Thus we saw that a smaller set of transformed predictor variables could suffice instead of the larger set of untransformed predictors. This concept is very similar to the idea behind the method of principal components. The method of Principal Components has two main objectives

(2.1) Objective 1 Provide a method of data reduction, or reducing the dimension of the problem.

Objective 2 Provide a way to better interpret the data.

In order to explain these ideas we will first describe *population principal components.*

Population Principal Components

Assume that the $p \times 1$ dimensional random vector **X** is distributed as $N_p(\boldsymbol{\mu}, \Sigma)$. Let

(2.2) $\left(\lambda_1, \mathbf{e}_1\right), \ldots, \left(\lambda_p, \mathbf{e}_p\right)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$.

be the eigenvalue/eigenvector pairs for Σ. Define the *population principal components* as

(2.3) $Y_1 = \mathbf{e}_1{}' \mathrm{X}, \ldots, Y_p = \mathbf{e}_p{}' \mathrm{X}$

and write the *vector of population principal components as*

(2.4) $\mathbf{Y}' = \left(Y_1, \ldots, Y_p\right)$.

These are called "population" principal components because the true covariance matrix is used to compute them.

We call

$$\begin{aligned} & Y_1 = \text{ First Population Principal Component} \\ & Y_2 = \text{ Second Population Principal Component} \\ (2.5) \quad & \vdots \\ & Y_p = p-th \text{ Population Principal Component} \end{aligned}$$

The properties of the population principal components is given in the next result

Result 3. Let $\mathbf{X} \sim N_p\left(\boldsymbol{\mu}, \boldsymbol{\Sigma}\right)$ and **Y** be the vector of population principal components. Then **Y** is distributed as a *p*-dimensional multivariate normal random vector, with

$$V\{\mathbf{Y}\} = \begin{pmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \lambda_p \end{pmatrix} \equiv diag\left(\lambda_1, \ldots, \lambda_p\right)$$

and

$$\sum_{i=1}^{p} V\{X_i\} = \sigma_{11} + \sigma_{22} + \cdots + \sigma_{pp} = \sum_{i=1}^{p} V\{Y_i\} = \lambda_1 + \lambda_2 + \cdots + \lambda_p$$

i.e. $tr\left(\boldsymbol{\Sigma}\right)$ is the sum of the eigenvalues.

Therefore we can think of the population principal components as a full rank linear transformation of the random variable **X** which results in a new set of variables which are uncorrelated. In addition, we know by Result 2, of the previous section, that the first principal component is given by the linear combination (of unit length) which has maximal variance. Likewise the subsequent principal components can be thought of as those linear combinations (all of unit length) which give maximum variance while being perpendicular to all previous linear combinations. In addition Result 3 says that the *total variance* of $V\{\mathbf{X}\}$ is equal to the *total variance* of $V\{\mathbf{Y}\}$, where *total variance* is a scalar *measure of the amount of variability* in a covariance matrix and is computed by taking the sum of its diagonal elements.

A useful quantity associated with the $k-th$ population principal component is the *proportion of total population variance explained*, which is given by

(2.6) Proportion Explained = $\dfrac{\lambda_k}{\sum_{i=1}^{p} \lambda_i}$

Another useful quantity for the $k-th$ population principal component is the cumulative *proportion of total population variance explained,* which is given by

(2.7) Cumulative Proportion Explained = $\dfrac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{p} \lambda_i}$

Quite often, most of the total population variance is explained by the first few principal components. This explains why principal components analysis is used for data (or dimension reduction) since you can justify using only the first few principal components instead of the entire set of variables.

In order to explain how principal components analysis can be used for data interpretation we need the following result.

Result 4. Let $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{Y}$ be the vector of population principal components. Then

$$Corr\{X_k, Y_i\} = e_{ki}\sqrt{\frac{\lambda_i}{\sigma_{kk}}}$$

where e_{ki} is the $i-th$ component of the $k-th$ eigenvector e_k.

These correlations are often plotted in the following way. Let

(2.8) $$r_{k1} = Corr\{X_k, Y_1\}, r_{k2} = Corr\{X_k, Y_2\}$$

and create the p set of points $(r_{11}, r_{12}), (r_{21}, r_{22}), ..., (r_{k1}, r_{k2})$ and graph the p points on a bivariate plot keeping track of which point corresponds to which component (i.e. 1 through p). Then you look for groupings of points in two dimensions. If (r_{11}, r_{12}) and (r_{21}, r_{22}) are close to each other, this means that X_1 and X_2 are about equally correlated with principal components 1 and 2. These bivariate plots can be done for other pairs of principal components other than just the first two. An example is given in the next section on *Sample Principal Components.*

Sample Principal Components

In practice the covariance matrix of a random vector is never known, so it must be estimated. Let $\mathrm{X}_1, ..., \mathrm{X}_n$ be an independent random sample from a $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution, and let $\bar{\mathbf{X}}$ and **S** be the sample mean and covariance matrix. Let

(2.8) $$(\hat{\lambda}_1, \hat{\mathbf{e}}_1), ..., (\hat{\lambda}_p, \hat{\mathbf{e}}_p) \text{ where } \hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p.$$

be the eigenvalue/eigenvector pairs for **S.** Define the *sample principal components as*

(2.3) $$\hat{Y}_1 = \hat{\mathbf{e}}_1'\mathbf{X}, ..., \hat{Y}_p = \hat{\mathbf{e}}_p'\mathbf{X}$$

and write the *vector of sample principal components* as

(2.4) $$\hat{\mathbf{Y}}' = (\hat{Y}_1, ..., \hat{Y}_p)$$

We call

(2.5) $$\begin{aligned} \hat{Y}_1 &= \text{ First Sample Principal Component} \\ \hat{Y}_2 &= \text{ Second Sample Principal Component} \\ &\vdots \\ \hat{Y}_p &= p-th \text{ Sample Principal Component} \end{aligned}$$

The proportion of sample variance explained, and cumulative sample variance explained are defined by analogy with the previous section, as are the correlations in Result 4. We next work an example to illustrate the concepts we have discussed.

Example 2. The NLSY administered the *Armed Services Vocational Aptitude Battery (ASVAB)* test to all of its respondents, who were paid $50 dollars for their participation. The ASVAB consists of a series of questions on ten areas each of which receives a score. The ten areas are given below, along with the variable name we will associate with it

X_1	General Science
X_2	Arithmetic Reasoning
X_3	Word Knowledge
X_4	Paragraph Comprehension
X_5	Numerical Operations
X_6	Coding Speed
X_7	Auto and Shop Information
X_8	Mathematics Knowledge
X_9	Mechanical Comprehension
X_{10}	Electronics Information

We have a sample of $n = 11{,}857$ respondents which we will assume are distributed as $N_{10}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ random vectors. We obtained the estimated mean as

$$\bar{\mathrm{X}}' = \begin{pmatrix} 14.38 & 15.88 & 23.64 & 9.99 & 31.81 & 42.33 & 12.66 & 12.05 & 12.68 & 10.23 \end{pmatrix}$$

and for the estimated covariance matrix we got

$$\begin{pmatrix} 27.04 & 27.23 & 36.05 & 13.71 & 33.27 & 43.82 & 19.38 & 22.21 & 19.35 & 17.27 \\ 27.23 & 51.64 & 43.79 & 18.22 & 52.26 & 65.17 & 22.99 & 36.02 & 26.14 & 21.00 \\ 36.05 & 43.79 & 71.14 & 25.22 & 61.72 & 83.82 & 27.97 & 35.49 & 28.22 & 26.08 \\ 13.71 & 18.22 & 25.22 & 13.41 & 26.64 & 36.83 & 10.12 & 14.91 & 10.93 & 9.87 \\ 33.27 & 52.26 & 61.72 & 26.64 & 131.24 & 137.21 & 23.32 & 44.02 & 26.74 & 22.89 \\ 43.82 & 65.17 & 83.82 & 36.83 & 137.21 & 277.59 & 29.79 & 54.79 & 34.96 & 29.65 \\ 19.38 & 22.99 & 27.97 & 10.12 & 23.32 & 29.79 & 30.34 & 15.71 & 21.64 & 17.97 \\ 22.21 & 36.02 & 35.49 & 14.91 & 44.02 & 54.79 & 15.71 & 37.74 & 19.74 & 15.92 \\ 19.35 & 26.14 & 28.22 & 10.93 & 26.74 & 34.96 & 21.64 & 19.74 & 27.63 & 16.95 \\ 17.27 & 21.00 & 26.08 & 9.87 & 22.89 & 29.65 & 17.97 & 15.92 & 16.95 & 18.66 \end{pmatrix}$$

Notice that the Numerical Operations and Coding Speed have relatively large variances compared to the other tests.

It is hard to take in this large amount of information, so it is natural to try principal components analysis to see if we ca reduce the problem, or try to understand it better. Below we have the results of a SAS run with the following code (assuming the dataset to analyze has been previously created) where the variable name *T* (for test) is used instead of *X*. The default method in `PROC FACTOR` is to fit principal components (PCs).

```
PROC FACTOR NFACTOR=10 EV COV PREPLOT;
     VAR T1 T2 T3 T4 T5 T6 T7 T8 T9 T10;
```

Begin Output From SAS

```
---------------------------------------------------------------------------
Initial Factor Method: Principal Components

                   Prior Communality Estimates: ONE
                    Eigenvalues of the Covariance Matrix:
                   Total= 686.434827  Average= 68.6434827

                         1          2          3          4          5
 Eigenvalue       477.7808    96.6424    44.1545    20.8408    19.1981
 Difference       381.1385    52.4879    23.3137     1.6427    11.3905
 Proportion         0.6960     0.1408     0.0643     0.0304     0.0280
 Cumulative         0.6960     0.8368     0.9011     0.9315     0.9595

                         6          7          8          9         10
 Eigenvalue         7.8077     6.4840     5.5485     4.3469     3.6311
 Difference         1.3237     0.9354     1.2016     0.7157
 Proportion         0.0114     0.0094     0.0081     0.0063     0.0053
 Cumulative         0.9708     0.9803     0.9884     0.9947     1.0000

           10  factors will be retained by the NFACTOR criterion.
```

Eigenvectors

	1	2	3	4	5
T1	0.17239	0.26989	0.12425	-0.11423	-0.08656
T2	0.25090	0.33399	0.00474	0.57944	0.02561
T3	0.30900	0.35914	0.13901	-0.41802	-0.63068
T4	0.12899	0.11130	0.02505	-0.06195	-0.19982
T5	0.45220	0.01661	-0.85419	-0.19111	0.16684
T6	0.70264	-0.60919	0.36089	0.04264	0.05049
T7	0.13224	0.31436	0.22692	-0.29132	0.54505
T8	0.20700	0.24385	-0.04547	0.57862	-0.16762
T9	0.14602	0.29249	0.17633	0.01091	0.41431
T10	0.12396	0.24294	0.14158	-0.12034	0.16418

	6	7	8	9	10
T1	0.23714	0.16351	0.76687	-0.43629	0.00791
T2	-0.68831	0.01130	0.10468	-0.03657	-0.05357
T3	-0.11280	-0.11700	-0.29269	-0.00605	-0.26759
T4	-0.04356	-0.05934	-0.03001	0.10144	0.95392
T5	0.01653	-0.01381	0.02097	0.00297	-0.01788
T6	0.00350	0.01205	0.00724	0.00118	-0.01905
T7	-0.06488	0.51086	-0.36807	-0.20995	0.07430
T8	0.63507	0.21992	-0.28182	0.03329	-0.02250
T9	0.20330	-0.79621	-0.06721	-0.10228	-0.00248
T10	0.07894	0.11094	0.30738	0.86159	0.09370

Initial Factor Method: Principal Components

Factor Pattern

	FACTOR1	FACTOR2	FACTOR3	FACTOR4	FACTOR5
T1	0.72465	0.51025	0.15878	-0.10028	-0.07294
T2	0.76313	0.45688	0.00438	0.36809	0.01562
T3	0.80079	0.41860	0.10952	-0.22626	-0.32763
T4	0.76986	0.29876	0.04544	-0.07723	-0.23907
T5	0.86278	0.01426	-0.49546	-0.07616	0.06381
T6	0.92182	-0.35945	0.14393	0.01168	0.01328
T7	0.52478	0.56105	0.27375	-0.24145	0.43357
T8	0.73655	0.39023	-0.04918	0.43000	-0.11956
T9	0.60722	0.54702	0.22291	0.00948	0.34536
T10	0.62722	0.55286	0.21778	-0.12717	0.16653

	FACTOR6	FACTOR7	FACTOR8	FACTOR9	FACTOR10
T1	0.12743	0.08007	0.34739	-0.17493	0.00290
T2	-0.26763	0.00401	0.03431	-0.01061	-0.01420
T3	-0.03737	-0.03532	-0.08174	-0.00150	-0.06045
T4	-0.03324	-0.04126	-0.01930	0.05775	0.49635
T5	0.00403	-0.00307	0.00431	0.00054	-0.00297
T6	0.00059	0.00184	0.00102	0.00015	-0.00218
T7	-0.03291	0.23616	-0.15740	-0.07947	0.02570
T8	0.28887	0.09116	-0.10806	0.01130	-0.00698
T9	0.10807	-0.38571	-0.03012	-0.04057	-0.00090

```
         T10        0.05106    0.06539    0.16761    0.41583  -0.04133



Initial Factor Method: Principal Components

                 Plot  of Factor Pattern for FACTOR1 and FACTOR2

                                    FACTOR1
                                        1
                            F         .9
                                        E
                                       .8           C
                                                D   H B
                                       .7              A
                                                         J
                                       .6                I

                                       .5                G

                                       .4

                                       .3

                                       .2
                                                                           F
                                       .1                                  A
                                                                           C
        -1  -.9-.8-.7-.6-.5-.4-.3-.2-.1  0 .1 .2 .3 .4 .5 .6 .7 .8 .9 1.0T O
                                      -.1                                  R
                                                                           2

                                      -.2
                                      -.3
                                      -.4
                                      -.5
                                      -.6
                                      -.7
                                      -.8
                                      -.9
                                       -1
T1      =A     T2      =B     T3      =C     T4      =D     T5      =E
T6      =F     T7      =G     T8      =H     T9      =I     T10     =J

--------------------------------------------------------------------

                                      End Output From SAS
```

From the output we can see that the first principal component explains almost 70% of the total sample variance, while over 90% of the total sample variance is explained by the first three principal components. SAS gives a FACTOR PATTERN as part of the output, which are the correlations based on the sample analog of Result 4 where rows correspond to k and columns correspond to i. The pairs of points corresponding to the first two principal components are plotted on the last page of output. The graph does not lend itself to a great deal of interpretation. There is a mild amount of grouping of the points J, I, G (Electronics Information, Mechanical Comprehension, Auto and Shop Information). F (Coding Speed), and E (Numerical Operations) seem to be distinct from the rest of the points. It is hard to tell anything else from this plot.

Principal component analysis is often used as a way to deal with multicollinearity. We discussed multicollinearity is SURV 615, and found that it caused us problems when the predictor variables in the regression model were nearly collinear. By using principal components on the predictor variables, and retaining only the first few principal components we can usually get rid of the collinearity problem and still retain a model with a substantial amount of explainability. Recall that a variance inflation factor (VIF) depends on the correlation of an X_k with the other X's. If principal components of the $\mathbf{X}$ matrix are used to predict a $\mathbf{Y}$, then all VIFs are 1 because PC's are orthogonal.

Principal Components Analysis of the Correlation Matrix

We have been analyzing the covariance matrix, but in many cases it is preferable to analyze the correlation matrix. Using the correlation matrix will give a different set of PCs than will the covariance matrix. Analyzing the correlation matrix is equivalent to analyzing the covariance matrix of standardized variables. Standardization can be important when variables are on very different scales such as the AFQT percentile score and the two SAT scores given as an example in Class #1. If some components have large variances relative to other components then creating principal components on the original covariance matrix will result in relatively uninteresting principal components, since the first several principal components will tend to reproduce the components with the largest variances. Therefore as a matter of practice it is a good idea to analyze the correlation matrix (or standardize the variable) unless the components of the vector or all on roughly an equivalent scale. In Example 2, all the variables were on roughly an equivalent scale, so we did not standardize.

3. Factor Analysis

Factor analysis is similar to principal components analysis in that it attempts to come up with a relatively simple explanation for the covariance structure for a potentially large dimensional problem. Factor analysis has a long history which we do not have time to go into. It has its basis as psychometric testing, and is often related to measuring intelligence. We will only give a brief overview of the subject. In particular we will only discuss the *Orthogonal Factor Model*, and will only discuss *Maximum Likelihood Estimation* for the *Orthogonal Factor Model*. In the orthogonal factor model, factors have zero covariance as was the case in PCA. For further information, see the references given in the first class.

We begin by construction. Let $\mathbf{X}$ be a p-dimensional normal random vector constructed in the following manner

(3.1) $$\begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_P \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} + \begin{pmatrix} l_{11} \\ l_{21} \\ \vdots \\ l_{p1} \end{pmatrix} f_1 + \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_p \end{pmatrix}$$

where $f \sim N(0,1)$ and independent of

(3.2) $$\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_p \end{pmatrix} \sim N_p \left(\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} \psi_1 & 0 & 0 & 0 \\ 0 & \psi_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \psi_p \end{pmatrix} \right)$$

and $l_{11}, l_{21}, \ldots, l_{p1}$ are unknown parameters (l stands for "loading"). In this form we call the random variable f_1 a *common factor* and refer to the random variables $e_1, \ldots, e_p$ as *specific factors*. "Common" means that f is a common random effect for all p X's. Then we can write

(3.3) $$\mathrm{X} \sim N_p(\mu, \Sigma)$$

where

(3.4) $\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix}$ and $\Sigma = \begin{pmatrix} l_{11}^2 + \psi_1 & l_{11} l_{21} & \cdots & l_{11} l_{p1} \\ l_{11} l_{22} & l_{22}^1 & \cdots & l_{21} l_{p1} \\ \vdots & \vdots & \ddots & \vdots \\ l_{11} l_{p1} & l_{21} l_{p1} & \cdots & l_{pp}^2 + \psi_p \end{pmatrix}$

This covariance matrix is written in a specific form that is less general than

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & & \\ \vdots & & \ddots & \\ \sigma_{p1} & \sigma_{p2} & & \sigma_{pp} \end{bmatrix}$$

In general a $p \times p$ covariance matrix has $\frac{1}{2} p(p+1)$ distinct unknown elements since the matrix is symmetric. The representation in (3.4) is a function of only $2p$ free parameters (p l's and p ψ's) which for large p is a large reduction in the number of unknowns. Expressions (3.1) to (3.4) describe the *single factor orthogonal factor model*. The *k-variable orthogonal factor model* is given by

(3.5) $$\begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} + \sum_{i=1}^{k} \begin{pmatrix} l_{1i} \\ l_{2i} \\ \vdots \\ l_{pi} \end{pmatrix} f_i + \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_p \end{pmatrix}$$

where $f_1, f_2, ..., f_k$ are all distributed as independent $N(0,1)$ random variables and independent of

(3.6) $$\begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_p \end{pmatrix} \sim N_p \left(\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} \psi_1 & 0 & 0 & 0 \\ 0 & \psi_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \psi_p \end{pmatrix} \right)$$

Again the random variables $f_1, f_2, ..., f_k$ are called the common factors and $e_1, ..., e_p$ as *specific factors.* This is an orthogonal factor model because the f_i's are independent. The f_i are not observed and are also called *latent* variables. Note that the f_i's are independent since the covariance matrix is diagonal. Similar to (3.3) and (3.4), we can write

(3.7) $$\mathrm{X} \sim N_p(\mu, \Sigma)$$

where

(3.8) $$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} \sum_{i=1}^{k} l_{1i}^2 + \psi_1 & \sum_{i=1}^{k} l_{1i} l_{2i} & \cdots & \sum_{i=1}^{k} l_{1i} l_{pi} \\ \sum_{i=1}^{k} l_{1i} l_{2i} & \sum_{i=1}^{k} l_{2i}^2 + \psi_2 & \cdots & \sum_{i=1}^{k} l_{2i} l_{pi} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^{k} l_{1i} l_{pi} & \sum_{i=1}^{k} l_{2i} l_{pi} & \cdots & \sum_{i=1}^{k} l_{pi}^2 + \psi_p \end{pmatrix}$$

Notice that we can write

(3.9) $$V\{X_1\} = \sum_{i=1}^{k} l_{1i}^2 + \psi_1$$

we refer to

(3.10) $\sum_{i=1}^{k} l_{1i}^2 =$ Communality for X_1 (i.e. this is the communal contribution of the factors to the variance of X_1).

$\psi_1 =$ Specific Variance for X_1

the communalities and specific variances for the other components are defined analogously.

The equations in (3.1) and (3.5) look like regression equations, but they involve the common factors which are considered *latent variables* which are *not observable*. The idea behind factor analysis is to try and see if the covariance matrix of **X** has a structure which is the same (or approximately) as that of an orthogonal factor model with as small a number of factors as possible. If it can be approximated by such a model then the idea is to estimate the factor loading coefficients l_{ij} and the specific variances.

While all this may seem difficult, it is actually not as easy as it may seem! It turns out that the loading coefficients l_{ij} are not uniquely defined. To attempt to make this more clear, define

(3.11) $$\mathbf{L} = \begin{pmatrix} l_{11} & l_{12} & \cdots & l_{1k} \\ l_{21} & l_{22} & \cdots & l_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ l_{p1} & l_{p2} & \cdots & l_{pk} \end{pmatrix} \qquad \mathbf{f} = \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_k \end{pmatrix} \qquad \mathbf{e} = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_p \end{pmatrix}$$

then

(3.12) $$\mathbf{X}=\boldsymbol{\mu}+\mathbf{Lf}+\mathbf{e}$$

$$V\{\mathbf{X}\}=\mathbf{LL}'+\boldsymbol{\Psi}$$

where

(3.13) $$\boldsymbol{\Psi}=diag\left(\psi_1,\ldots,\psi_p\right)$$

Now let $\mathbf{Q}$ be an $k\times k$ orthogonal matrix and define

(3.14) $$\tilde{\mathbf{L}}=\mathbf{LQ}$$

then

(3.15) $$\tilde{\mathbf{L}}\tilde{\mathbf{L}}'=\mathbf{LQQ}'\mathbf{L}'=\mathbf{LL}'$$

so in terms of the covariance matrix $V\{\mathbf{X}\}$ the matrices $\mathbf{L}$ and $\tilde{\mathbf{L}}$ are indistinguishable. In other words, the model $\mathbf{X}=\boldsymbol{\mu}+\mathbf{Lf}+\mathbf{e}$ is equivalent to the model $\mathbf{X}=\boldsymbol{\mu}+\tilde{\mathbf{L}}\mathbf{f}+\mathbf{e}$.

It turns out that by placing additional restrictions on the model, the method of maximum likelihood can produce unique estimators of the factor loading coefficients and the specific variances. These extra restrictions imply that there are fewer free parameters in the general *k*-factor model than you may think. It turns out you can show the following result

Result 5. Let $\mathbf{X}\sim N_p\left(\boldsymbol{\mu},\boldsymbol{\Sigma}\right)$ and assume we want to fit a *k*-factor model then

1) $\boldsymbol{\Sigma}$ has $\frac{1}{2}p\left(p+1\right)$ distinct elements
2) The factor model has $p\left(k+1\right)-\frac{1}{2}k\left(k-1\right)$ free parameters
3) Leaving $\frac{1}{2}\left[\left(p-k\right)^2-p-k\right]$ degrees-of-freedom

Next we give possible factor model configurations for $p\le 10$. Note that if a configuration is not listed then it corresponds to a model with *negative-degrees-of-freedom* which means that it does not correspond to a valid model.

Table 1. Factor Model Configurations for $p \le 10$.

p	k	$\frac{1}{2}p(p+1)$	*free*	$df = \frac{1}{2}\left[(p-k)^2 - p - k\right]$
3	1	6	6	0
4	1	10	8	2
5	1	15	10	5
5	2	15	14	1
6	1	21	12	9
6	2	21	17	4
6	3	21	21	0
7	1	28	14	14
7	2	28	20	8
7	3	28	25	3
8	1	36	16	20
8	2	36	23	13
8	3	36	29	7
8	4	36	34	2
9	1	45	18	27
9	2	45	26	19
9	3	45	33	12
9	4	45	39	6
9	5	45	44	1
10	1	55	20	35
10	2	55	29	26
10	3	55	37	18
10	4	55	44	11
10	5	55	50	5
10	6	55	55	0

A nice feature of likelihood estimation of the factor model is that it can test the hypothesis of the form

$$H_0 : k = k_0 \qquad versus \quad H_A : k > k_0 \tag{3.16}$$

where k_0 is a specified number of factors. This is done using what is called a *likelihood ratio test.* We do not have time to go into the details of these tests, but both SAS and SPSS will provide these tests as part of the output along with significance levels for the tests.

Once the unknown parameters of the problems have been estimated (i.e. factor loadings and specific variances) it is common to further *rotate the factor loadings* to help in interpreting the solutions. There are many ways to automatically choose such a rotation, but one that is particularly popular is called *varimax.* The varimax option is available in SAS and SPSS.

Once the factor loadings have been rotated we can look at bivariate factor loading plots similar to those we did in principal components analysis. We illustrate this next with an example.

Example 3. Johnson and Wichern (2002, pp. 502-504) give a small example of rotation using data from Lawley and Maxwell (1971). Tests were given to 220 male students on 6 subjects. The estimated factor loadings were:

Table 9.5 from Johnson and Wichern (2002)

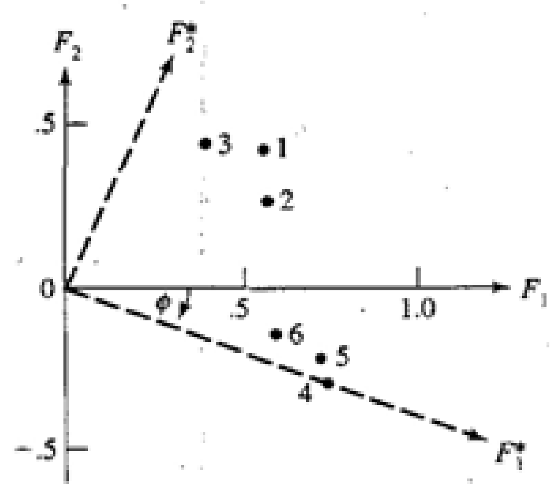
	Estimated factor loadings	
Variable	F_1	F_2
1. Gaelic	0.553	0.429
2. English	0.568	0.288
3. History	0.392	0.450
4. Arithmetic	0.740	-0.273
5. Algebra	0.724	-0.211
6. Geometry	0.595	-0.132

All the variables have positive loadings on the first factor. Lawley and Maxwell suggest this is a *general intelligence* factor. The second factor is hard to interpret since the mathematical subjects have negative loading but the other subjects (Gaelic, English, History) have positive loadings. Factors with somewhat clearer interpretations can be obtained by rotation.

Quoting Johnson and Wichern:

"The factor loading pairs $\left(\hat{\ell}_{i1}, \hat{\ell}_{i2}\right)$ are plotted as points in Figure 9.1. The points are labeled with the numbers of the corresponding variables. Also shown is a clockwise orthogonal rotation of the coordinate axes through an angle of $\phi = 20^\circ$. This angle was chosen so that one of the new axes passes through $\left(\hat{\ell}_{41}, \hat{\ell}_{42}\right)$. When this is done, all the points fall in the first quadrant (the factor loadings are all positive), and the two distinct clusters of variables are more clearly revealed.

The mathematical test variables load highly on F_1^* and have negligible loadings on F_2^*. The first factor might be called a *mathematical-ability* factor. Similarly, the three verbal test variables have high loadings on F_2^* and moderate to small loadings on F_1^*. The second factor might be labeled a *verbal-ability* factor. The *general-intelligence* factor identified initially is submerged in the factors F_1^* and F_2^*.



Example 4. We continue the analysis begun in Example 2 with the ASVAB data taken from NLSY. We created the SAS output with the code, again with the variable name *T* (for test) instead of *X*.

```
PROC FACTOR N=4 METHOD=ML COV ROTATE=VARIMAX PREPLOT PLOT;
     VAR T1 T2 T3 T4 T5 T6 T7 T8 T9 T10;
```

Begin Output From SAS

```
------------------------------------------------------------------------
-- Initial Factor Method: Maximum Likelihood

                   Prior Communality Estimates: SMC

                    T1        T2        T3        T4        T5
              0.773522  0.761549  0.799137  0.709161  0.626355

                    T6        T7        T8        T9       T10
              0.559430  0.665842  0.712296  0.690312  0.726751

     Preliminary Eigenvalues:  Total = 25.329264  Average = 2.5329264

                       1           2           3           4           5
 Eigenvalue      23.3602      2.3059      0.8626      0.4449     -0.0829
 Difference      21.0543      1.4433      0.4177      0.5278      0.1166
 Proportion       0.9223      0.0910      0.0341      0.0176     -0.0033
 Cumulative       0.9223      1.0133      1.0474      1.0649      1.0616

                       6           7           8           9          10
 Eigenvalue      -0.1995     -0.2725     -0.3468     -0.3605     -0.3822
 Difference       0.0731      0.0743      0.0137      0.0217
 Proportion      -0.0079     -0.0108     -0.0137     -0.0142     -0.0151
 Cumulative       1.0538      1.0430      1.0293      1.0151      1.0000


           4 factors will be retained by the NFACTOR criterion.

Iter Criterion    Ridge   Change   Communalities

  1    0.01682    0.000  0.15243   0.80157 0.81695 0.91125 0.73734 0.74139
                                   0.71186 0.79930 0.84736 0.75189 0.77680

  2    0.01585    0.000  0.01235   0.80245 0.81564 0.92026 0.74297 0.75103
                                   0.69951 0.79791 0.85506 0.75407 0.77500

  3    0.01584    0.000  0.00239   0.80242 0.81511 0.92077 0.74300 0.74935
                                   0.70190 0.79794 0.85622 0.75388 0.77491

  4    0.01584    0.000  0.00010   0.80241 0.81504 0.92084 0.74300 0.74942
                                   0.70182 0.79784 0.85632 0.75390 0.77491


                    Convergence criterion satisfied.

Initial Factor Method: Maximum Likelihood
```

```
Significance tests based on 11857 observations:

   Test of H0: No common factors.
        vs HA: At least one common factor.

   Chi-square = 106400.05   df = 45   Prob>chi**2 = 0.0001

   Test of H0: 4 Factors are sufficient.
        vs HA: More factors are needed.

   Chi-square = 187.716   df = 11   Prob>chi**2 = 0.0001

Chi-square without Bartlett's correction = 187.82393064
Akaike's Information Criterion = 165.82393064
Schwarz's Bayesian Criterion = 84.636520053
Tucker and Lewis's Reliability Coefficient = 0.9932026948
```

Squared Canonical Correlations

FACTOR1	FACTOR2	FACTOR3	FACTOR4
0.973844	0.794054	0.702443	0.565151

Eigenvalues of the Weighted Reduced Correlation Matrix:
Total = 44.7487873 Average = 4.47487873

	1	2	3	4	5
Eigenvalue	37.2328	3.8556	2.3607	1.2996	0.1195
Difference	33.3772	1.4949	1.0610	1.1802	0.0597
Proportion	0.8320	0.0862	0.0528	0.0290	0.0027
Cumulative	0.8320	0.9182	0.9710	1.0000	1.0027

	6	7	8	9	10
Eigenvalue	0.0597	0.0166	-0.0369	-0.0767	-0.0822
Difference	0.0431	0.0535	0.0397	0.0056	
Proportion	0.0013	0.0004	-0.0008	-0.0017	-0.0018
Cumulative	1.0040	1.0044	1.0036	1.0018	1.0000

Factor Pattern

	FACTOR1	FACTOR2	FACTOR3	FACTOR4
T1	0.88255	-0.12363	-0.05694	-0.07065
T2	0.85833	0.05967	0.27087	-0.03730
T3	0.91474	0.06304	-0.26307	-0.10446
T4	0.83243	0.16131	-0.15016	-0.03872
T5	0.71237	0.39269	0.02886	0.29478
T6	0.64880	0.39446	-0.06115	0.34863
T7	0.71487	-0.50317	0.01054	0.18307
T8	0.81981	0.19313	0.35219	-0.15133
T9	0.77651	-0.34031	0.16131	0.09541
T10	0.81462	-0.33135	0.00521	0.03859

Initial Factor Method: Maximum Likelihood

Variance explained by each factor

	FACTOR1	FACTOR2	FACTOR3	FACTOR4
Weighted	37.232798	3.855645	2.360696	1.299649
Unweighted	6.423412	0.874727	0.323138	0.294243

Final Communality Estimates and Variable Weights
Total Communality: Weighted = 44.748787   Unweighted = 7.915519

	T1	T2	T3	T4	T5
Communality	0.802415	0.815044	0.920840	0.743003	0.749415
Weight	5.061073	5.406704	12.632622	3.891086	3.990678

	T6	T7	T8	T9	T10
Communality	0.701820	0.797844	0.856324	0.753900	0.774915
Weight	3.353671	4.946682	6.960087	4.063420	4.442763

```
Initial Factor Method: Maximum Likelihood

              Plot of Factor Pattern for FACTOR1 and FACTOR2

                                 FACTOR1
                                    1

                                A  .9C
                                     B   D
                          IJ       .8    H

                    G              .7          E
                                               F
                                   .6

                                   .5

                                   .4

                                   .3

                                   .2
                                                                    F
                                   .1                               A
                                                                    C
     -1 -.9-.8-.7-.6-.5-.4-.3-.2-.1  0 .1 .2 .3 .4 .5 .6 .7 .8 .9 1.0T
                                                                    O
                                  -.1                               R
                                                                    2
                                  -.2

                                  -.3

                                  -.4

                                  -.5

                                  -.6

                                  -.7

                                  -.8

                                  -.9

                                   -1

T1      =A    T2      =B    T3      =C    T4      =D    T5       =E
T6      =F    T7      =G    T8      =H    T9      =I    T10      =J
```

Rotation Method: Varimax

Orthogonal Transformation Matrix

	1	2	3	4
1	0.58908	0.48098	0.44384	0.47397
2	-0.75859	0.59550	0.23764	0.11599
3	0.05920	-0.08604	0.73276	-0.67243
4	0.27207	0.63768	-0.45783	-0.55654

Rotated Factor Pattern

	FACTOR1	FACTOR2	FACTOR3	FACTOR4
T1	0.59109	0.31071	0.35295	0.48158
T2	0.46625	0.40127	0.61069	0.25236
T3	0.44704	0.43353	0.27603	0.67591
T4	0.34858	0.48466	0.31549	0.53578
T5	0.20367	0.76198	0.29569	0.19973
T6	0.17420	0.77453	0.17728	0.20036
T7	0.85325	0.16004	0.12162	0.17149
T8	0.31611	0.38252	0.73711	0.25836
T9	0.75109	0.21779	0.33829	0.16700
T10	0.74205	0.21866	0.26896	0.32270

Variance explained by each factor

	FACTOR1	FACTOR2	FACTOR3	FACTOR4
Weighted	15.243781	10.526647	9.092169	9.886191
Unweighted	2.902681	2.128099	1.537001	1.347739

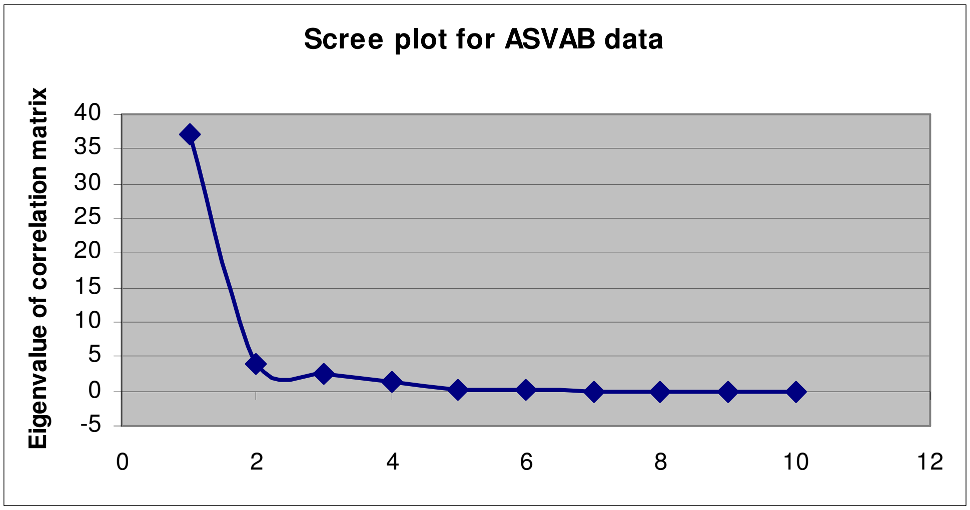
Final Communality Estimates and Variable Weights
Total Communality: Weighted = 44.748787   Unweighted = 7.915519

	T1	T2	T3	T4	T5
Communality	0.802415	0.815044	0.920840	0.743003	0.749415
Weight	5.061073	5.406704	12.632622	3.891086	3.990678

	T6	T7	T8	T9	T10
Communality	0.701820	0.797844	0.856324	0.753900	0.774915
Weight	3.353671	4.946682	6.960087	4.063420	4.442763

```
Rotation Method: Varimax

              Plot of Factor Pattern for FACTOR1 and FACTOR2

                                  FACTOR1
                                     1

                                    .9
                                         G
                                    .8
                                           I
                                    .7

                                    .6        A

                                    .5
                                                 BC
                                    .4
                                                   D
                                    .3          H

                                    .2                     E
                                                            F         F
                                    .1                                A
                                                                      C
     -1  -.9-.8-.7-.6-.5-.4-.3-.2-.1  0 .1 .2 .3 .4 .5 .6 .7 .8 .9 1.0T
                                                                      O
                                   -.1                                R
                                                                      2
                                   -.2

                                   -.3

                                   -.4

                                   -.5

                                   -.6

                                   -.7

                                   -.8

                                   -.9

                                    -1

T1      =A    T2      =B    T3      =C    T4      =D    T5      =E
T6      =F    T7      =G    T8      =H    T9      =I    T10     =I
```

One method of determining the number of factors to use is called a *score plot.* The sorted eigenvalues under the part of the output labeled “Eigenvalues of the Weighted Reduced Correlation Matrix” are plotted versus a counter (1,2,…,10).



The idea is to look for the point where the plot flattens out near zero. In this case, 4 factors seem more than adequate even though the test of H_0 : (4 factors are sufficient) is rejected. Remember, when the sample size is large, statistically significant results can be found that are not practically significant.

It is interesting that the hypothesis of a four factor model is rejected by the data. In their book The Bell Curve by Richard Herrnstein and Charles Murray argue that the data are consistent with a single factor model where the single factor represents underlying intelligence. Actually in the book they correct for the effect of age which we did do not here, but eh basic model is the same. If we look at the pre-rotation graph we see some slight groupings of the component test: 1) G,I,J (Auto and Shop Information, Mechanical Comprehension, Electronics Information); 2) A (General Science); 3) B,C,D,H (Arithmetic Reasoning, Word Comprehension, Paragraph Comprehension, Mathematics Knowledge); 4) E,F (Numerical Operations, Coding Speed). If we look at the varimax rotation graph we see the same patterns but they appear more clearly. Interestingly the third and fourth groups which we isolated (Arithmetic Reasoning, Word Comprehension, Paragraph Comprehension, Mathematics Knowledge) make up the components which go into the AFQT which we looked at before, and is used in The Bell Curve as the basis for their measure of intelligence.

Example 5. Another method of interpreting rotated factors is given in this example. The Defense Manpower Data Center (DMDC) in the Department of Defense conducts surveys of military personnel to determine job satisfaction. The surveys include many different items relating to job assignments, workload, how well a person likes their co-workers and supervisor, how satisfied they are with pay, quality of equipment and many other aspects of military life. Based on one personnel survey, JPSM faculty performed a factor analysis and obtained the varimax rotated loadings shown in the following table.

One approach to identifying factors that you can give a meaningful name to is to first sort by loadings for factor 1. Identify the variables with absolute values of the loadings ≥ 0.50. These are marked in bold in the column under factor 1. Then sort the absolute values of the factor 2 loadings for the remaining factors and mark the ones whose values are ≥ 0.50. These are shown in bold in column 2. In this case, we used only three factors. The final step was to sort the variables not used in factors 1 and 2 by the loadings for the third factor, giving the three bolded variables under factor 3.

We, thus, obtain three factors that were named *Social/work satisfaction, Organizational commitment, and Satisfaction with Equipment, and Facilities.* Given these factors and the loadings for the variables assigned to each, we might try to predict which service people were more or less likely to re-enlist in the service when their next opportunity arose. In fact, one of the important questions on the DMDC surveys is the so-called “retention intention” question that asks just that.

VARIMAX ROTATED LOADINGS		Factors		
		1	2	3
Question	**Social/Work satisfaction**			
RC080A	type of assignments	**0.72**	0.21	0.27
RC080B	stability of assignments	**0.68**	0.19	0.27
RC080C	workload	**0.66**	0.15	0.20
RC076B	enjoyment	**0.64**	0.47	0.11
RC080D	time required	**0.64**	0.26	0.20
RC076C	morale	**0.63**	0.22	0.24
RC076D	opportunity for leadership	**0.61**	0.24	0.20
RC075D	happy with life in NG/R	**0.59**	0.59	0.11
RC079A	training	**0.55**	0.12	0.49
SATNGRE	supervisor	**0.55**	0.15	0.20
SATNGRB	work	**0.54**	0.24	0.14
RC075A	enjoy serving	**0.54**	0.58	0.05
RC076E	opportunity to use skills	**0.54**	0.18	0.36
SATNGRD	coworkers	**0.52**	0.13	0.18
	Organizational commitment (Attitudes toward serving)			
RC075I	if I left, would feel like I let country down	0.12	**0.71**	0.11
RC075C	would feel guilty if I left	0.16	**0.69**	0.10
RC075F	have sense of obligation	0.27	**0.67**	0.11
RC075K	helps me achieve what I want	0.38	**0.64**	0.10
RC075E	difficult to give up benefits	0.22	**0.63**	0.17
RC075J	leaving would require sacrifice	0.10	**0.61**	0.16
RC075B	consistent with goals	0.47	**0.61**	0.07
RC075G	military values are my own	0.34	**0.60**	0.04
RC075H	proud to be in NG/R	0.45	**0.57**	0.03
RC075L	intend to leave at next opportunity	-0.32	**-0.56**	-0.08
	Satisfaction with Equipment, Facilities			
RC079C	equipment	0.32	0.11	**0.83**
RC079D	mechanical condition	0.29	0.10	**0.77**
RC079B	facilities	0.40	0.10	**0.58**
	Less important variables			
SATNGRA	compensation	0.34	0.23	0.16
SATNGRC	promotion	0.45	0.12	0.12
RC080E	likelihood of activation	0.40	0.46	0.12
RC080G	no. of recent activations	0.39	0.35	0.11
RC076A	military values	0.50	0.47	0.10
RC080F	likelihood of no activation	0.05	-0.14	0.02